Project Design Phase

Proposed Solution Template

Date: 01 October 2026

Team ID: SWTID-2026-8643

Project Name: Script-Controlled ACL - Restrict Record Access Based on Field Value

Maximum Marks: 2 Marks

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	In ServiceNow, users who can access a table may normally view its records. For an institution, this can expose student details such as email, phone number and faculty information across ECE, EEE and CSE to users who only require their own branch data. There is also no clear separation between viewing, creating, editing and deleting records. The institution therefore needs record-level security based on the Branch field and the user's role.
2	Idea / Solution description	Implement script-controlled Access Control Lists (ACLs) on the custom table **u_institution_details**. Use separate custom roles for each operation: bb1 for Read, bb2 for Create, bb3 for Write and bb4 for Delete. The Read ACL combines the Branch = EEE condition with an advanced script: administrators receive full access, bb1 users see only EEE records, and other users are denied. Validate the setup by impersonating an EEE user, a user without the roles and an administrator.
3	Novelty / Uniqueness	The solution combines a field-based condition with an advanced script in one ACL, so access depends on both the user's role and the record's content. Each operation uses a separate role, supporting the least-privilege principle. Administrator override is retained, separate tables for each branch are unnecessary, and the design uses native ServiceNow features without external tools or plugins.
4	Social Impact / Customer Satisfaction	The solution protects student personal information and privacy while improving trust between students, faculty and the institution. It reduces the possibility of accidental or intentional misuse and supports privacy and audit requirements. Branch staff receive a focused view containing only relevant records, while administrators retain complete control.
5	Business Model (Revenue Model)	This is an internal ServiceNow security capability, so its main value comes from reducing compliance and data-breach risk rather than immediate sales. It could be offered as a reusable ACL template or update set for educational institutions, included in ServiceNow implementation and consulting services, or delivered as a managed security-configuration service.
6	Scalability of the Solution	Other branches such as ECE and CSE can be added through additional roles and data conditions without redesigning the overall approach. The same pattern can protect other tables containing sensitive information. Roles may be assigned to groups instead of individual users, and the ACL configuration can be packaged as a scoped application or update set for movement between instances. Server-side ACL evaluation supports large record volumes.